

Full Name: _____
EEL 3135 – Classwork #05

Milestone:

- Complete any 2 questions to receive credit.

Question #1: Given:

$$x(t) = 4e^{j(10t+\pi/6)} + 4e^{-j(10t+\pi/6)}$$

- (a) Use Euler's identity to convert $x(t)$ into its real sinusoidal form.
- (b) Identify the amplitude, phase shift, and frequency of the resulting cosine wave.
- (c) Sketch one period of the waveform in Matlab.

Question #2: You are given the following frequency-domain information for $y(t)$:

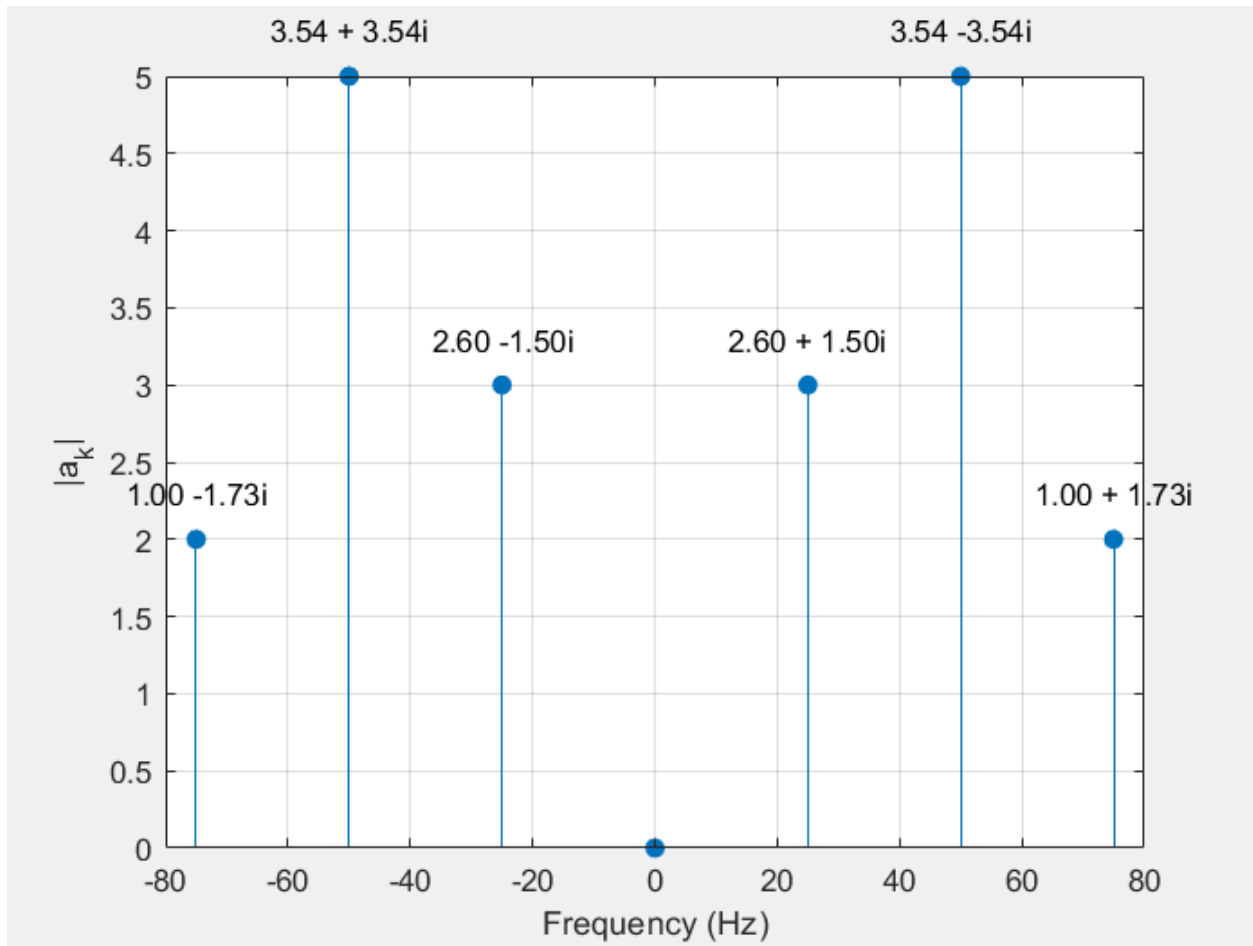
$$\omega_0 = 5 \text{ rad/s}, \quad a_0 = 3, \quad a_{\pm 1} = 2e^{\mp j\pi/4}, \quad a_{\pm 2} = 1e^{\mp j\pi/3}$$

(a) Write $y(t)$ as a sum of complex exponentials.

(b) Rewrite $y(t)$ using cosine terms (real Fourier series).

(c) Plot $y(t)$ over one period in MATLAB.

Question #3: The figure below shows the two-sided magnitude spectrum of $x(t)$ with frequency components at $f = \pm 25, \pm 50, \pm 75$ Hz. Each spectral line is labeled with its corresponding complex coefficient a_k in rectangular form (real + imaginary).



(a) Determine the fundamental frequency f_0 and fundamental period T_0 of $x(t)$.

(b) Write $x(t)$ in complex exponential form using the coefficients a_k from the spectrum.

(c) Use Euler's formula to rewrite $x(t)$ as a sum of cosines in the time domain.

Question #4: Given

$$y(t) = 4 \cos\left(12t + \frac{\pi}{3}\right) - 7 \cos\left(24t - \frac{\pi}{6}\right) + 2 \cos(36t),$$

(a) Evaluate $y\left(\frac{\pi}{12}\right)$.

(b) Express $y(t)$ in complex exponential form using Euler's formula.

(c) Sketch the magnitude spectrum of $y(t)$ at the harmonic frequencies $k\omega_0$ using Matlab.